

Structural Consistency Is Not Retrieval Utility: An Exact-and-Heuristic Audit of Preference-Graph Repair for Multi-Ranker Retrieval

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Abstract

Purpose: [Stage 2 drafting. State the research question: whether enforcing acyclicity in derived multi-ranker preference graphs – via heuristic or exact minimum-weight feedback-arc-set repair – yields a statistically supported improvement in retrieval effectiveness, and what that implies about using structural consistency as a proxy for retrieval utility.]

Methods: [Stage 2 drafting. Summarize the audited pipeline (construction, repair, extraction, evaluation), the four-benchmark canonical package, the three vote-construction regimes, greedy vs. exact SCIP repair, and the Holm-corrected paired-inference protocol.]

Results: [Stage 2 drafting. State the central null/conditional result with the headline Holm-corrected cell counts and the exact-repair confirmation, without overstating effect direction.]

Conclusion: [Stage 2 drafting. State the restrained conclusion: structural repair is real and measurable, but is not a reliable surrogate for downstream retrieval utility; implications for practitioners and future preference-graph studies.]

Keywords: preference graphs, feedback arc set, minimum-weight feedback arc set, graph repair, rank aggregation, information retrieval, retrieval evaluation, data quality

1 Introduction

Pairwise preferences are a natural and increasingly common form of ranking evidence. Rather than committing to a single scalar relevance score, a pairwise judgment states

only that one document should be ranked ahead of another for a given query. This form of evidence arises whenever several ranking signals are available for the same query – lexical and dense retrievers, cross-encoders, or large language models used as pointwise, pairwise, or listwise judges – and it arises directly whenever a system elicits pairwise comparisons instead of absolute scores. Aggregating such judgments across several rankers or several repeated comparisons for the same query produces a natural representation: a query-specific weighted directed preference graph, in which an edge $u \rightarrow v$ records support for ranking document u ahead of document v , and the edge weight records how strongly that support should count.

That representation is attractive because it preserves disagreement rather than collapsing it. It is also structurally fragile. Because the graph is built from several partially disagreeing sources, it can contain directed cycles – $u \rightarrow v \rightarrow w \rightarrow u$ – so that no single total order is consistent with every retained edge. Cycles of this kind are not a pathological edge case; they are the expected outcome whenever comparison evidence is aggregated across imperfectly correlated sources. Because a cycle is a case where the evidence contradicts itself, it is natural to treat acyclicity, or a reduction in cyclic and contradictory structure, as a desirable property of the graph, and cycle repair (removing or reversing a minimum-weight set of conflicting edges) as an attractive preprocessing step before a ranking is read off the graph. This intuition is plausible on its face: a graph that no longer contradicts itself seems like a strictly better input to whatever ranking rule is applied to it next.

That plausibility, however, rests on treating four distinct stages of a preference-graph pipeline as if they were one. *Construction* decides which pairwise votes are retained and how they are weighted, and is itself sensitive to score normalization, vote semantics, and thresholding choices made before any graph exists. *Structural repair* operates on the resulting graph and optimizes a purely graph-internal objective – the total weight of edges that must be removed or reversed to restore acyclicity. *Ranking extraction* reads a final ordering off the (possibly repaired) graph using some scoring rule. *Downstream evaluation* then compares that ordering against relevance judgments using a retrieval metric that has no direct relationship to the repair objective. Repair minimizes disagreement with the graph’s own edges; retrieval metrics score agreement with an external notion of relevance. Nothing in the construction of the repair objective guarantees that reducing the former improves the latter, and nothing in the standard presentation of repair work makes that gap explicit.

This gap is the question prior work leaves open: does improving a preference graph’s structural consistency – as measured by the repair objective itself – also improve the retrieval effectiveness of the ranking eventually read off that graph? Existing evidence does not answer this cleanly, for three reasons that compound. First, essentially all empirical evaluations of preference-graph repair use heuristic repair methods, so any observed retrieval effect, positive or null, is open to the objection that it reflects the heuristic’s suboptimality rather than a property of repair itself: a null result could always be dismissed as “the heuristic simply did not repair enough.” Second, graph construction is rarely held explicit and varied alongside repair, so it is difficult to attribute an observed effect (or its absence) specifically to the repair

step rather than to upstream choices that determine how much contradictory structure the graph contained in the first place. Third, studies that report both structural diagnostics (cyclicity, removed edge weight) and a retrieval metric do not generally test whether the two move together under corrected statistical inference; an improvement in the former is often reported alongside, and implicitly read as evidence for, the latter.

This paper addresses that gap directly. We hold graph construction explicit across three vote-construction regimes that vary how much contradictory structure a graph retains, and we compare, on the same underlying candidate sets and ranker scores, *unrepaired*, *greedy-repaired*, and *exactly repaired* preference graphs. The greedy-only design that dominates prior work cannot, by itself, resolve the central question: a null retrieval result under a heuristic repair method is always consistent with the hypothesis that a better repair would have revealed a gain the heuristic missed. We therefore include exact minimum-weight feedback-arc-set (MWFAS) repair, solved to certified optimality with an open-source mixed-integer solver, as a diagnostic control rather than as a proposed practical method: if exact repair, which by construction cannot be out-performed by any heuristic on the repair objective itself, still fails to produce a reliable retrieval gain, then heuristic suboptimality is ruled out as the explanation for a null result. This is what exact repair is for in this study – not a faster or better production repair algorithm, but a diagnostic instrument that isolates whether the retrieval question has an answer the heuristic could plausibly have obscured.

The paper’s scope is stated honestly rather than maximally. The main empirical setting is score-derived multi-ranker retrieval: preference graphs built from stored BM25, TF-IDF, and MiniLM scores across four public benchmarks spanning distinct retrieval domains. A smaller real-large-language-model pairwise-judgment pilot is included only as a bounded, directional addendum, not as a second confirmatory study; it is not treated as validating or generalizing the main findings. The paper does not claim state-of-the-art retrieval effectiveness, does not propose a new minimum-weight feedback-arc-set formulation or solver, does not claim that repair never helps retrieval in any setting, does not claim that exact repair universally fails to help, and does not offer a validated criterion for deciding, per query, when repair is worth applying.

Within that scope, the central finding is restrained rather than sweeping. Structural repair is genuinely active: under the permissive construction regime, both greedy and exact repair remove a measurable share of total graph weight, and evaluating against a candidate pool larger than the metric cutoff shows that repair can change which documents even enter the evaluated top- k set. What does not follow is a reliable retrieval benefit. Across four benchmarks, three construction regimes, and both repair methods, no repaired-versus-unrepaired normalized discounted cumulative gain (nDCG) comparison survives Holm-corrected paired significance testing. The paper’s thesis is therefore that structural graph consistency and downstream retrieval utility are related but empirically distinct quality dimensions, and that enforcing acyclicity should not be treated as a validated surrogate for improving retrieval effectiveness without evidence obtained under the retrieval metric itself.

This paper is a rigorous empirical and computational study of that question, not a proposal for a new or superior reranking method. Its contributions are:

1. A controlled empirical framework that separates preference-graph construction, structural repair, ranking extraction, and downstream retrieval evaluation into explicit, independently varied stages, so that any observed effect can be attributed to a specific stage rather than to their combination.
2. A three-way comparison, under identical construction and evaluation protocols, of unrepaired, greedy-repaired, and exactly (SCIP-based MWFAS) repaired preference graphs – using exact repair as a diagnostic control that closes the standard objection that a null retrieval result merely reflects heuristic under-repair.
3. A four-benchmark, multiplicity-corrected empirical finding: graph repair is structurally active and can change which documents are evaluated at all, yet no repaired-versus-unrepaired nDCG family – canonical, larger-pool, or exact-repair – survives Holm-corrected paired testing, so structural consistency is not shown to be a reliable general surrogate for retrieval utility.
4. An open, reproducible implementation and evidence pipeline covering graph construction, greedy and exact repair, ranking extraction, and the statistical evaluation protocol, so that the audit reported here can be rerun, checked, or extended by others.

2 Related Work

2.1 Pairwise Ranking and Preference Aggregation

Representing evidence as pairwise comparisons rather than absolute scores is classical in ranking theory. Given comparisons $u \succ v$, two largely separate traditions convert them into a ranking. The first is *probabilistic preference modeling*: Bradley–Terry-type models explain observed comparison outcomes through latent per-item strengths estimated by maximum likelihood [3, 16], and Plackett–Luce models generalize the same idea to full or partial orderings [26, 31]. Rank Centrality recovers a ranking as the stationary distribution of a Markov chain built from pairwise comparisons, without fitting a parametric strength model [28], and combinatorial Hodge theory decomposes a pairwise comparison graph into a gradient (globally consistent) component and a curl/harmonic (locally or globally cyclic) residual, giving a spectral account of how far the data are from admitting a single ranking [19, 39]. Bias-aware pairwise ranking further shows that even the sampling process that generates which pairs are compared can itself introduce systematic ranking bias, independent of any repair step [12].

The second tradition is *combinatorial rank aggregation and cycle repair*, which treats a set of pairwise preferences (or several ranked lists) as combinatorial input and asks for an ordering, or a modification of the input, that minimizes a specific disagreement objective. Kemeny’s optimal-consensus ranking minimizes total pairwise disagreement with a set of input rankings [21]; web-scale rank aggregation work popularized Markov-chain, Copeland-style (wins-minus-losses), and Borda-style (positional) aggregation rules and analyzed their properties [8]; and follow-up work gave approximation algorithms and complexity results for Kemeny-optimal and related

aggregation problems, connecting them to correlation clustering [1, 22]. Rank comparison metrics that tolerate ties and partial overlap, such as the Fagin–Kumar family, are a companion line of work for measuring how much two rankings (or a ranking before and after a repair step) actually differ [10, 11].

These two traditions answer different questions. Probabilistic models estimate a ranking that best explains noisy comparison outcomes and treat residual disagreement as statistical noise around a latent order. Combinatorial repair instead treats the observed graph as fixed evidence and asks how to modify it, at minimum cost, so that a single consistent order becomes possible. The present paper is positioned entirely in the second tradition: it uses combinatorial minimum-weight feedback-arc-set repair (Section 3.4) on an already-constructed preference graph, and its research question is not about estimating latent item strengths but about whether the combinatorial repair objective, once optimized, is diagnostic of a different objective – retrieval effectiveness – that it was never designed to optimize.

2.2 Pairwise and Graph-Based Ranking in Information Retrieval

Pairwise comparison has become a common building block of modern retrieval and reranking pipelines. Pairwise ranking prompting elicits document-level comparisons directly from a large language model and aggregates them into a ranking [33], and pointwise, pairwise, and listwise LLM reranking paradigms have been compared directly for effectiveness and cost [36]. Listwise and setwise rerankers reduce the number of language-model calls needed to reorder a candidate list [14, 32, 43, 47], and hybrid rerankers combine lexical, dense, and cross-encoder signals within a single scoring or fusion step [25, 29, 41, 45]. Graph-free score and rank fusion – reciprocal rank fusion, CombSUM, and Borda-style combination – remains a strong and widely used alternative to any graph-based aggregation step, in part because it avoids comparing incompatible raw score magnitudes [2, 4, 6, 13, 24, 27, 40].

A smaller body of work builds an explicit preference graph from pairwise document judgments before extracting a ranking, rather than fusing scores or ranks directly. **LLM-RankFusion** aggregates several LLM rankers by explicitly modeling and mitigating pairwise inconsistency in the resulting comparisons [44], and confidence- or bias-aware pairwise aggregation methods weight or correct individual comparisons before they are combined [12]. This is the setting the present paper studies: a query-specific weighted directed graph built from several ranking signals, from which a ranking is extracted before or after a structural repair step. Standard retrieval benchmarking practice – reusing stored candidate pools and score files rather than re-running upstream retrieval, and evaluating with rank-sensitive metrics such as nDCG – follows the same evaluation conventions established for zero-shot retrieval benchmarking more generally [7, 37].

2.3 Inconsistency, Cycles, and Preference Repair

A pairwise preference graph is transitive, and therefore admits a single consistent total order, only in the special case that it is acyclic. In general it is not: aggregating several

partially disagreeing sources routinely produces directed cycles, including simple two-way contradictions and longer cyclic chains. The combinatorial problem of removing the least-weight set of edges needed to make a weighted directed graph acyclic is the (minimum-weight) feedback arc set problem. It is NP-hard in general [20]; classical work gives fast, deterministic greedy heuristics that repeatedly remove the lightest edge on a discovered cycle [9], and later work gives polynomial-time approximation schemes and connects the problem to correlation clustering [1, 22]. This paper does not claim the minimum-weight feedback-arc-set formulation itself as a contribution – it is a long-studied classical combinatorial optimization problem – and instead studies the empirical consequence of solving it, both heuristically and to certified optimality, for a downstream retrieval task.

The same structural issue has recently reappeared in the context of large language models used as pairwise or comparative judges: repeated or multi-evaluator LLM comparisons frequently produce transitivity violations and directed cycles, which several recent studies diagnose or attempt to resolve [15, 35, 46]. That work generally treats acyclicity (or reduced contradiction) as a desirable property of the resulting evaluation and reports downstream gains on tasks such as model ranking, response selection, or fine-tuning-data selection [15]; it does not test whether the acyclicity objective itself is diagnostic of retrieval effectiveness under a corrected, paired statistical protocol, which is the specific gap this paper addresses.

This paper substantially extends and revises an earlier, unpublished feedback-arc-set repair investigation by the same author, which studied a smaller and partially overlapping evidence base under a less conservative evaluation protocol (no separation of canonical from larger-pool evaluation cells, no Holm correction applied to the headline retrieval claim, and no exact-repair comparison) and reported a conditional, dataset-dependent positive retrieval effect. That manuscript did not complete peer review and is not cited here as a prior publication. The present study is not a resubmission of it: it uses a materially more conservative statistical protocol, adds exact SCIP-based repair specifically to rule out heuristic-repair artifacts as an explanation, and reaches a different, more restrained conclusion on an updated evidence base. Readers should treat the two as distinct, non-comparable analyses rather than as an earlier and later version of the same result.

2.4 Structural Quality Versus Downstream Utility

A recurring pattern across the work reviewed above is that structural or intrinsic quality – cyclicity, removed feedback-arc weight, agreement with a probabilistic comparison model, or acyclicity of an aggregated evaluation graph – is measured and reported, but rarely tested directly against an independent downstream task metric under corrected statistical inference. This mirrors a broader distinction in the data-quality literature between *intrinsic* quality (properties of the data or artifact itself) and *contextual* or task-relative quality (fitness for a specific use) [42]; work on data cascades and benchmark label error shows concretely that upstream curation or construction choices can shape downstream conclusions in ways that are easy to miss when only the final task metric is inspected [30, 34]. Simple significance testing is also known to be an unreliable guide to retrieval system comparisons unless multiplicity

and pairing are handled correctly [5, 17, 38], and the interpretation of a non-significant result requires care distinct from ordinary significance testing [23].

Few studies in the preference-graph or feedback-arc-set literature isolate structural improvement from downstream task utility as two separately measured and separately tested quantities. Most either report only structural diagnostics, or report a task metric without a repair-method control (e.g. exact repair) that would rule out heuristic suboptimality as the explanation for a null or weak downstream effect. This paper’s contribution relative to this literature is exactly that controlled separation: it holds graph construction explicit, compares unrepaired, heuristically repaired, and exactly repaired graphs on the same evidence, and tests the structural-repair-to-retrieval-utility link directly under a multiplicity-corrected paired protocol, rather than assuming that a gain on one objective implies a gain on the other.

3 Background and Problem Formulation

This section fixes notation and formalizes the four stages distinguished in Section 1 – retrieval setting, derived preference graph, structural inconsistency, and repair – before turning to ranking extraction and evaluation. Every definition below is stated at the level of generality needed to state the paper’s research question precisely; the specific benchmarks, ranker families, and evaluated cells that instantiate this formulation are given in Section 4.

3.1 Retrieval and Candidate Rankings

Let $q \in \mathcal{Q}$ denote a query drawn from a fixed query set, and let $D_q = \{d_1, \dots, d_n\}$ denote its query-specific candidate document set: the pool of documents that will be compared and ranked for q . Candidate sets are formed upstream of this study’s pipeline, typically as the union of documents retrieved or scored by one or more base rankers, and are held fixed once formed.

A base ranker (or, more generally, a judgment source) $s \in S$ assigns a score $x_{q,s}(d) \in \mathbb{R}$ to each document $d \in D_q$ it scores; a ranker need not score every document in D_q , so scores may be missing for some (q, s, d) triples. Because different rankers’ raw scores are not generally comparable in scale or distribution, scores are normalized per query and per ranker before they are compared across rankers. The normalization used throughout the canonical experiments is per-query, per-ranker min-max scaling,

$$\bar{x}_{q,s}(d) = \frac{x_{q,s}(d) - \min_{d' \in D_q} x_{q,s}(d')}{\max_{d' \in D_q} x_{q,s}(d') - \min_{d' \in D_q} x_{q,s}(d')}, \quad (1)$$

which maps every ranker’s retained scores for a given query into a common $[0, 1]$ range. Section 3.2 builds pairwise votes from $\bar{x}_{q,s}$ rather than from $x_{q,s}$ directly; the raw, unnormalized construction is retained only as an explicit robustness ablation, not as an alternative primary protocol, precisely because raw heterogeneous score margins are not a scale-neutral starting point for comparing rankers.

Relevance judgments (qrels) associate each query-document pair with a graded or binary relevance label, drawn from each benchmark’s original human-annotated

judgments; they are external to, and never derived from, any ranker score or preference graph. All retrieval metrics reported in this paper are computed against these external judgments, evaluated at a fixed prefix length k (the evaluation cutoff): a metric such as $\text{nDCG}@k$ scores only the top- k documents of a candidate ranking. Because the candidate pool D_q evaluated by a ranking method may be larger than the cutoff k , the relationship between the pool size $P = |D_q|$ (or a truncated evaluation pool size) and k is itself an explicit design choice returned to in Section 4: repair can only change which documents occupy the top- k set when $P > k$ gives it room to do so.

3.2 Derived Weighted Preference Graph

For each query q , the normalized ranker scores $\bar{x}_{q,s}$ are converted into a weighted directed preference graph

$$G_q = (V_q, E_q, w_q), \quad V_q = D_q, \quad (2)$$

in which a directed edge $(u, v) \in E_q$ records retained evidence that document u should be ranked ahead of document v , and the edge weight $w_q(u, v) > 0$ records the strength of that retained evidence. The graph is a *derived* data object, not ground truth: it is the output of an explicit construction procedure applied to the normalized scores, and its structure – which edges exist, in which direction, and with what weight – depends on that procedure’s choices, not on any property of the documents themselves.

Construction proceeds ranker by ranker and pair by pair. For each ranker s and each unordered document pair $\{u, v\} \subseteq D_q$ that s scores, s casts a directional vote for whichever of $u \succ v$ or $v \succ u$ its normalized scores support, with vote strength derived from the normalized score margin $|\bar{x}_{q,s}(u) - \bar{x}_{q,s}(v)|$. A pair with an exactly equal normalized score, or for which s does not score both documents, abstains rather than casting an imputed vote: ties and missing comparisons are treated as an absence of evidence, not as evidence of indifference. Votes across rankers are then aggregated per directed pair, and a pair-level and graph-level retention rule (the *vote-construction regime*) decides which directions are actually retained as edges of G_q .

This paper studies three explicit vote-construction regimes that hold the candidate set and underlying ranker scores fixed and vary only the retention rule:

- **ms2** (conservative): retain a direction only when it has majority-style support across rankers and the aggregated margin exceeds a fixed threshold. This suppresses most direct two-way contradictions and, in the canonical package, produces graphs that are nearly always acyclic.
- **ms1** (permissive): retain every ranker-supported directional vote. Because different rankers may disagree, both (u, v) and (v, u) can be retained for the same unordered pair, producing direct mutual contradictions and, transitively, longer cycles.
- **ms1_drop_mutual**: start from **ms1** and remove every unordered pair that would otherwise be retained in both directions, before aggregation. This isolates how much of **ms1**’s cyclicity is attributable specifically to direct mutual contradiction rather than to longer-cycle nontransitivity.

These three regimes are not alternative proposals for a "correct" construction rule; they are a controlled sweep whose purpose is to make graph construction, rather than the repair algorithm, a first-class experimental factor (Section 4 reports how strongly this factor determines observed cyclicity). Because a conflicting vote-construction regime can produce a graph in which repair has little or no structural work to do, an observed repair effect (or its absence) cannot be interpreted without knowing which regime produced the graph it was applied to.

3.3 Cycles and Structural Inconsistency

A weighted directed graph G_q is *acyclic* if it contains no directed cycle $u_1 \rightarrow u_2 \rightarrow \dots \rightarrow u_m \rightarrow u_1$; equivalently, its strongly connected components (maximal vertex sets within which every vertex is reachable from every other) are all singletons. G_q admits at least one linear extension consistent with every retained edge direction if and only if it is acyclic; a cyclic G_q cannot be represented exactly by any single total order over D_q ; some retained edge must be violated by any linear ranking of the graph's vertices.

This paper reports two purpose-specific structural inconsistency measures, rather than a single omnibus cyclicity score. The first is a per-query binary indicator of whether G_q is cyclic at all under a given construction regime, aggregated across queries into a cyclic-query rate. The second decomposes that rate into cyclicity attributable to direct mutual contradiction (a pair retained in both directions) versus cyclicity that survives after mutual pairs are removed (comparing `ms1` to `ms1_drop_mutual`), because these two sources of cyclicity have different plausible causes – near-tied evidence between two rankers versus genuine longer-range nontransitivity among three or more documents – and conflating them would over- or under-state how much of the observed cyclicity a repair step is actually resolving.

Acyclicity is a purely structural property: it guarantees that G_q admits *some* consistent total order, but it does not by itself guarantee that the retained edges are correct, that the graph's implied ranking agrees with relevance judgments, or that any particular extraction rule applied to an acyclic graph will outperform one applied to a cyclic graph. Transitivity, acyclicity, evidential correctness, and retrieval optimality are four distinct properties; this paper treats none of them as substitutes for any other, and Sections 3.4 and 3.5 keep the objective that repair optimizes (feedback-arc weight) explicitly separate from the metric evaluation uses (nDCG against qrels).

3.4 Minimum-Weight Feedback-Arc-Set Repair

Given a (possibly cyclic) preference graph $G_q = (V_q, E_q, w_q)$, repair seeks a minimum-weight set of edges $F_q^* \subseteq E_q$ whose removal leaves an acyclic graph:

$$F_q^* = \arg \min_{F \subseteq E_q} \sum_{e \in F} w_q(e) \quad \text{s.t.} \quad (V_q, E_q \setminus F) \text{ is acyclic.} \quad (3)$$

This is the minimum-weight feedback arc set (MWFAS) problem (Section 2.3); the repaired graph is $\tilde{G}_q = (V_q, E_q \setminus F_q^*, w_q)$. The objective in Equation (3) is entirely graph-internal: it minimizes disagreement with G_q 's own retained edges, subject to

the feasibility condition that the result be acyclic. It contains no reference to relevance judgments, ranking metrics, or any notion of retrieval utility – a property that is central to this paper’s research question rather than an incidental modeling detail.

Two solution methods for Equation (3) are used and reported as co-equal, not as a primary method with a secondary check. **Greedy repair** is a fast, deterministic heuristic: while $\tilde{G}_q$ contains a directed cycle, find one such cycle and remove its minimum-weight edge, repeating until the graph is acyclic. This does not generally reach the optimum of Equation (3), but it is cheap enough to run at the scale of the full query set. **Exact repair** solves Equation (3) to certified optimality via a standard linear-ordering mixed-integer program: for every ordered pair of nodes (u, v) , a binary variable $\text{before}[u, v]$ indicates that u precedes v in a chosen total order, subject to antisymmetry ($\text{before}[u, v] + \text{before}[v, u] = 1$) and transitivity constraints over every ordered triple of nodes; the objective minimizes the total weight of edges whose direction disagrees with the chosen order,

$$\min \sum_{(u,v) \in E_q} w_q(u, v) \text{before}[v, u], \quad (4)$$

which is exactly Equation (3) restricted to orderings representable as a linear extension. This program is solved with the open-source SCIP solver; a query graph is included in the exact-repair evidence only when the solver certifies proven optimality, not merely a feasible incumbent.

Exact repair is used here as a *diagnostic control*, not as a proposed practical replacement for greedy repair: it does not scale to the largest query graphs, and this paper makes no claim about the general scalability or practicality of exact MWFAS solving. Its role is narrower and specific to this study’s question. Because F_q^* from exact repair is, by construction, a global optimum of Equation (3), no repair method can remove less weight while still restoring acyclicity. If exact repair’s downstream retrieval effect is still statistically indistinguishable from greedy repair’s, then the suboptimality of the greedy heuristic is ruled out as an explanation for a null or weak retrieval effect – precisely the objection a greedy-only study cannot itself close (Section 1).

3.5 Ranking Extraction and Evaluation

A final query-specific ranking is extracted from either the unrepaired graph G_q or the repaired graph $\tilde{G}_q$, or from the upstream ranker signal alone. The canonical experiments compare four families of extraction rule, evaluated in both unrepaired and repaired form where the rule is graph-dependent:

- **Prior-only:** rank by the upstream per-document score (e.g. a normalized score sum across rankers), independent of any graph.
- **Graph-free fusion:** reciprocal rank fusion, CombSUM, and Borda-style fusion over the constituent rankers’ individual rankings, which likewise does not construct or consult a preference graph.

- **Graph-only extraction:** a Copeland-style score $s_{\text{cop}}(u) = \deg^+(u) - \deg^-(u)$ (unweighted out-degree minus in-degree); a weighted balance score $s_{\text{bal}}(u) = \sum_{(u,v) \in E} w(u,v) - \sum_{(v,u) \in E} w(v,u)$ (weighted out-edge mass minus in-edge mass); and a Rank-Centrality-style stationary-distribution score computed from a Markov chain over the (weighted) graph. Each of these three rules is computed on both G_q (unrepaired) and $\tilde{G}_q$ (repaired), holding the extraction rule fixed so that any difference is attributable to repair rather than to a change of scoring rule.
- **Hybrid prior-plus-graph extraction:** the graph score is combined with the normalized prior score,

$$h_q(d) = (1 - \alpha) \hat{s}_q^{\text{prior}}(d) + \alpha \hat{s}_q^{\text{graph}}(d), \quad (5)$$

with both components normalized query-wise and $\alpha \in [0, 1]$ fixed in the primary analysis (with a sweep reported as a robustness check). When $\hat{s}_q^{\text{graph}}$ is computed from G_q the method is unrepaired; when computed from $\tilde{G}_q$ it is repaired.

Retrieval effectiveness is measured with normalized discounted cumulative gain (nDCG), the standard graded-relevance ranking metric [18]: for a ranking with relevance labels $\text{rel}_1, \dots, \text{rel}_k$ at ranks $1, \dots, k$, $\text{DCG}@k = \sum_{i=1}^k (2^{\text{rel}_i} - 1) / \log_2(i+1)$, and $\text{nDCG}@k = \text{DCG}@k / \text{IDCG}@k$, where $\text{IDCG}@k$ is the DCG of the ideal (qrels-sorted) ranking at the same cutoff. nDCG is computed per query and compared between the unrepaired and repaired variant of the same extraction rule, holding every other choice fixed; Section 4 specifies the paired, multiplicity-corrected inference protocol used to test whether an observed repaired-versus-unrepaired difference is reliable.

The formulation above makes explicit why improving Equation (3) is not, by construction, the same as improving nDCG. The repair objective is computed entirely from G_q 's own edges and weights, which are themselves a derived function of upstream normalization, vote semantics, and thresholding choices (Section 3.2); nDCG is computed from an extraction rule's output ranking against relevance judgments that never enter the repair objective at all. A smaller feedback-arc weight, or proven-optimal removal of that weight, is evidence about internal consistency of the retained votes – it is not, on its own, evidence about agreement with relevance judgments. Whether the two are empirically linked in this study's evidence, and how reliably, is the question Sections 4 and 5 take up directly.

4 Experimental Design

5 Results

6 Discussion

7 Limitations

8 Conclusion

9 Data Availability and Reproducibility

Declarations

Funding

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Conflict of interest

The author declares no conflict of interest.

Data and code availability

The datasets used in this study are publicly available from their original sources and are cited in the manuscript. The code and processed artifacts used to generate the reported results are available at <https://github.com/SoroushVahidi/consistency-aware-llm-rankin>.

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